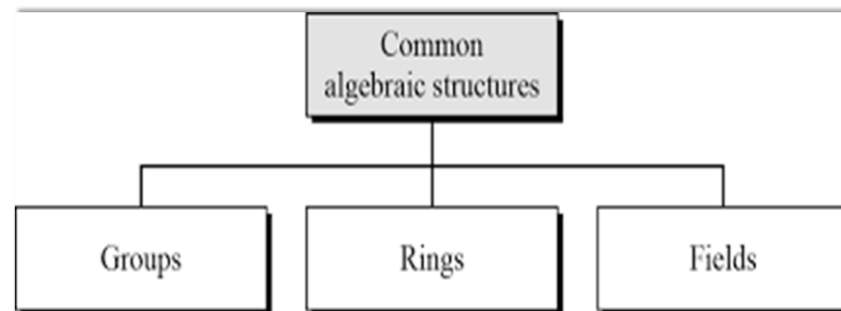


ALGEBRA 1

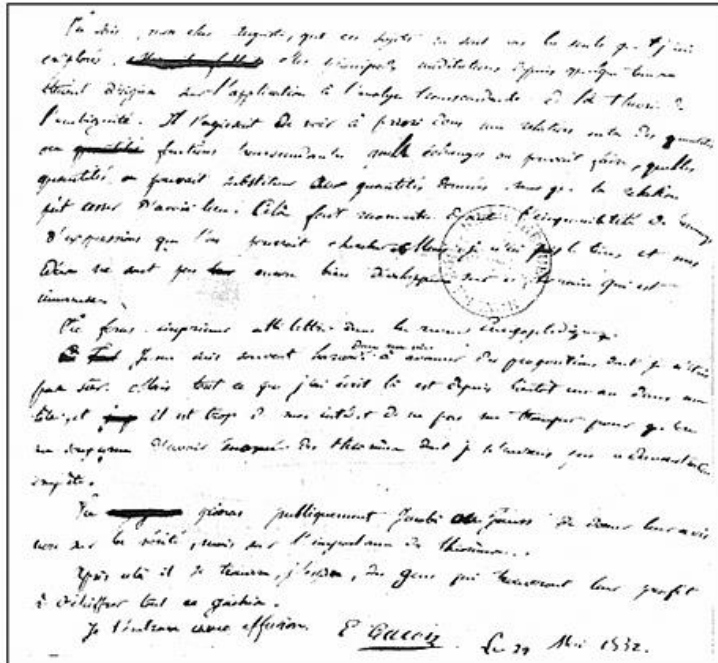
Chapter 4

ALGEBRAIC STRUCTURES

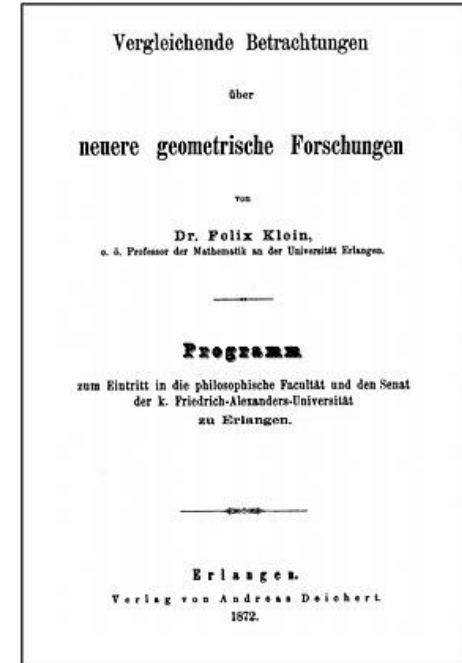
Main algebraic structures studied in this
1st semester



Precursors



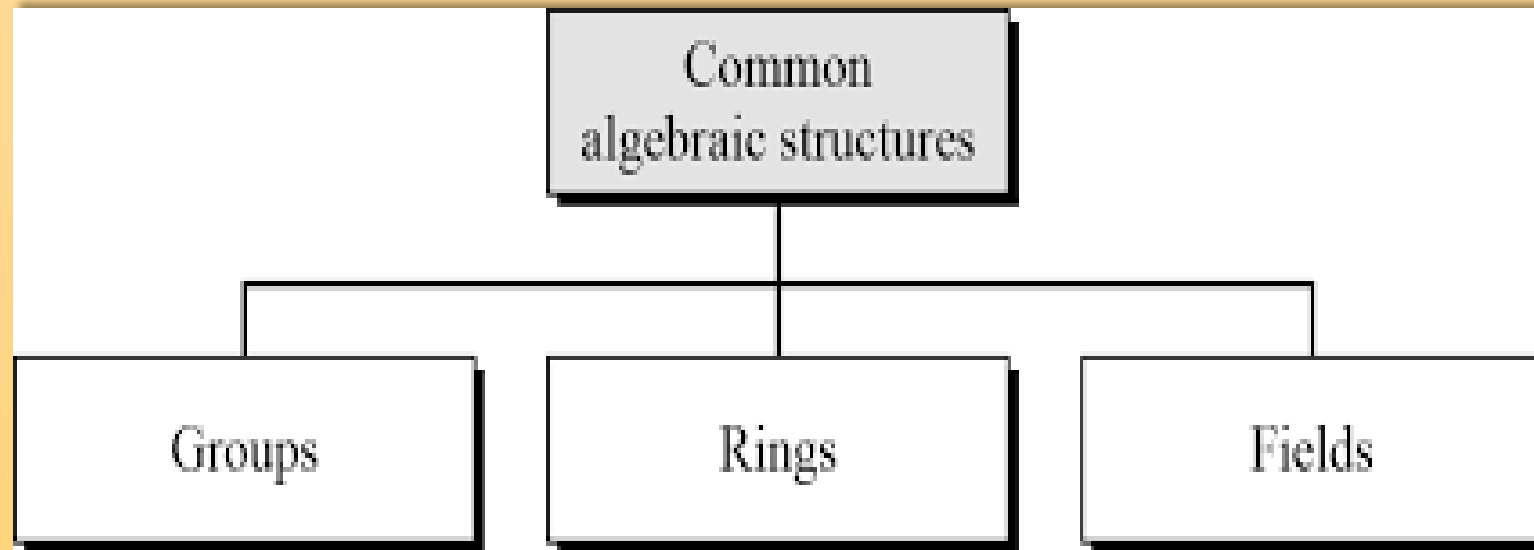
E. Galois



F. Klein

Algebraic structures (1st semester)

- An algebraic structure is a set (called carrier set or underlying set) with one or more binary operations defined on it that satisfies a list of axioms. Examples of algebraic structures include groups, rings, fields.



Topic List of some applications of Group Theory



- Logic/boolean algebra: hardware design, testing, artificial intelligence, software engineering
- Mathematical reasoning/induction: algorithm design, programming languages
- Number theory/probability: cryptography, security, algorithm design, machine learning
- Relations/relational algebra: databases
- Graph theory: networking, social networks, optimization

Binary Operation

Definition: A binary operation on a set S is a function

$$\diamond : S \times S \longrightarrow S$$

$$(x, y) \mapsto \diamond (x, y) = x \diamond y$$

Example:

The function $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(x, y) = xy + y$$

is a binary operation on $\mathbb{N}$. (*Check it ...*)

What is Group Theory?

Group theory is a branch of abstract algebra that studies the algebraic structures known as groups.

A group is a set equipped with a single [binary operation](#) that satisfies certain axioms.

Group theory has profound applications in various fields, including physics, chemistry, computer science, and cryptography.

Understanding the basic properties and concepts of groups provides a foundation for exploring more complex algebraic structures and their applications.

Group Definition

A set G together with a binary operation $*$ (often called multiplication or addition) is called a **group** if it satisfies the following four properties:

Closure:

For every pair of elements a and b in G , the result of the operation $\mathbf{a * b}$ is also in G , i.e.

$$\forall \mathbf{a, b \in G, a * b \in G}$$

Associativity:

For every three elements a , b , and c in G , the equality $\mathbf{(a * b) * c = a * (b * c)}$ holds, i.e.

$$\forall \mathbf{a, b, c \in G, (a * b) * c = a * (b * c)}$$

Identity Element:

There exists an element e in G (called the identity element) such that for every element a in G , the equations $\mathbf{e * a = a}$ and $\mathbf{a * e = a}$ hold, i.e.

$$\exists \mathbf{e \in G \ \forall a \in G, (e * a = a) \wedge (a * e = a)}$$

Inverses:

For every element a in G , there exists an element b in G (called the inverse of a , often denoted $\mathbf{a^{-1}}$) such that

$$\mathbf{a * b = e \text{ and } b * a = e, \text{ i.e.}}$$

$$\forall \mathbf{a \in G, \exists b \in G \ a * b = e \text{ and } b * a = e}$$

Examples of Group.

Some of the important examples of groups are discussed below:

Example #1:

- **Integers under Addition ($\mathbb{Z}, +$)**
- **Set:** $\mathbb{Z}$, the set of all integers: $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
- **Operation:** Addition (+).
- **Identity Element:** 0 (since $n + 0 = n$ for any integer n).
- **Inverse Element:** For any integer n , the inverse is $-n$ (since $n + (-n) = 0$).

Example #2:

- **Non-Zero Real Numbers under Multiplication ($\mathbb{R}^*, \cdot$)**
- **Set:** $\mathbb{R}^*$, the set of all non-zero real numbers.
- **Operation:** Multiplication ($\cdot$).
- **Identity Element:** 1 (since $n \cdot 1 = n$ for any non-zero real number n).
- **Inverse Element:** For any non-zero real number n , the inverse is $1/n$ (since $n \cdot 1/n = 1$).

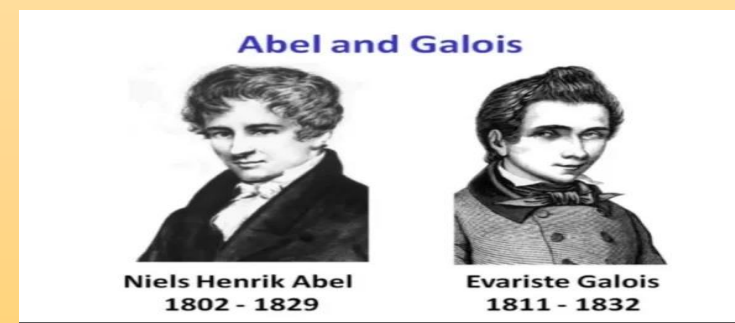
Other Algebraic Structure Related to Group

Abelian Group

Semi Group

Monoid

Let's discuss these in detail as follows:



- **Abelian Group or Commutative group**

- For a non-empty set S , $(S, *)$ is called a Abelian group if it follows the following axioms:
- **Closure:** $(a * b)$ belongs to S for all $a, b \in S$.
- **Associativity:** $a * (b * c) = (a * b) * c$ where a, b, c belongs to S .
- **Identity Element:** There exists $e \in S$ such that $a * e = e * a = a$ where $a \in S$.
- **Inverses:** For $a \in S$ there exists $a^{-1} \in S$ such that $a * a^{-1} = a^{-1} * a = e$.
- **Commutative:** $a * b = b * a$ for all $a, b \in S$.

Note:

Every abelian group is a group, monoid, semigroup, and algebraic structure,

Semi Group

A non-empty set S , $(S,*)$ is called a semigroup if it follows the following axiom:

- **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
- **Associativity:** $a*(b*c) = (a*b)*c$ where a, b, c belongs to S .

Note: A semi-group is always an algebraic structure.

Example: (Set of natural numbers, +), and $(\mathbb{Z}, *)$ are examples of semigroup.

Monoid

A non-empty set S , $(S,*)$ is called a monoid if it follows the following axiom:

- **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
- **Associativity:** $a*(b*c) = (a*b)*c$, $\forall a, b, c \in S$.
- **Identity Element:** There exists $e \in S$ such that $a*e = e*a = a$, $\forall a \in S$

Note: A monoid is always a semi-group and algebraic structure.

Examples of Various Algebraic Structures

Magma : C.

Semi Group : C.A.

Monoid :C.A.I.

- Specifically, a **magma** consists of a set equipped with a single binary operation that must be closed by definition. No other properties are imposed.
- $(\mathbb{Z}, \times)$ is **monoid** as 1 is an integer which is also an identity element.
- $(\mathbb{N}^*, +)$ is not Monoid as there doesn't exist any identity element. But this is Semigroup.
- But $(\mathbb{Z}, +)$ is Monoid with 0 as identity element.

Here is a Table with different nonempty set and operation

N: Set of [Natural Number](#), (without 0)

Z: Set of [Integers](#),

R: Set of [Real Numbers](#),

E: Set of [Even Numbers](#), (or $2\mathbb{Z}$)

Set, Operation	Algebraic Structure	Semi Group	Monoid	Group	Abelian Group
N, +	Y	Y	–	–	–
N, –	–	–	–	–	–
N, ×	Y	Y	Y	–	–
N, ÷	–	–	–	–	–
Z, +	Y	Y	Y	Y	Y
Z, –	Y	–	–	–	–
Z, ×	Y	Y	Y	–	–
Z, ÷	–	–	–	–	–
R, +	Y	Y	Y	Y	Y
R, –	Y	–	–	–	–
R, ×	Y	Y	Y	–	–
R, ÷	–	–	–	–	–
E, +	Y	Y	Y	Y	Y
E, ×	Y	Y	–	–	–

WORKED EXAMPLE 1: (multiplicative notation)

Prove that in a group G , if $(ab)^2 = a^2b^2$ for all $a, b \in G$, then G is abelian.

- *Given: $(ab)^2 = a^2b^2$ for all $a, b \in G$*
- *Expand $(ab)^2$: $abab = a^2b^2$*
- *Multiply both sides by a^{-1} on the left: $a^{-1}(abab) = a^{-1}(a^2b^2)$*
- *Simplify: $bab = ab^2$*
- *Multiply both sides by b^{-1} on the right: $(bab)b^{-1} = (ab^2)b^{-1}$*
- *Simplify: $ba = ab$*
- *Therefore, G is abelian.*

Worked Example2 : (multiplicative notation)

Let G be a group and let $a, b \in G$. Prove that if $ab = ba$, then $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$.

- *We'll use induction on n .*
- Base case: For $n = 0$, $(ab)^0 = e = a^0 b^0$
For $n = 1$, $(ab)^1 = ab = a^1 b^1$
- Inductive hypothesis: Assume $(ab)^k = a^k b^k$ for some $k \geq 1$
- Inductive step: Consider $(ab)^{k+1}$
- $(ab)^{k+1} = (ab)^k(ab) = (a^k b^k)(ab) = a^k(b^k a)b = a^k(ab^k)b = a^{k+1}b^{k+1}$
- By induction, $(ab)^n = a^n b^n$ for all $n \geq 0$
- For negative integers, note that $(ab)^{-n} = ((ab)^n)^{-1} = (a^n b^n)^{-1} = b^{-n} a^{-n} = a^{-n} b^{-n}$
- Therefore, $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$.

Subgroup of a group (1/5)

(notations : $H < G$ or $H \leq G$)

Definition: If a subset H of a group G is itself a group under the operation of G , we say that H is a subgroup of G .

Subgroup (2/5)

Ostensibly, to check that a subset H is a subgroup, we would need to check all four properties of the group. That is, closure (ie, the operation gives a map $H \times H \rightarrow H$; products of things in H are always in H), identity, the inverse, and associativity. In fact, since H has the same operation as G , we know that the operation in H is associative (since G is a group). Furthermore, if the operation is closed and inverses exist, then we know that for any $h \in H$, h^{-1} must be in H .

So really we only need to check two things:

1. Closure: $gh \in H$ for all $g, h \in H$, and
2. Inverses: $h^{-1} \in H$ for all $h \in H$.

Some important things to notice:

1. The group G is always a subgroup of itself!
(G is a subset of itself, which is a group with the same operation as G .)
2. The subset containing just the identity element is also a subgroup! This is called the trivial subgroup.

Subgroup Tests

One-Step Subgroup Test

Theorem: Let G be a group and H a nonempty subset of G . Then, H is a subgroup of G if ab^{-1} is in H whenever a and b are in H . (In additive notation, H is a subgroup if $a - b$ is in H whenever a and b are in H .)

Proof.

1. **Associativity:** Since the operation of H is the same as that of G , associativity holds.
2. **Identity:** Since H is nonempty, we may pick some x in H . Let $a = x$ and $b = x$. Then $ab^{-1} = xx^{-1} = e$.
3. **Inverses:** Let $a = e$ and $b = x$. Then, $ab^{-1} = ex^{-1} = x^{-1}$.
4. **Closure:** Assume x, y are in H . We have shown that y^{-1} is in H . Let $a = x$ and $b = y^{-1}$. Then $xy = x(y^{-1})^{-1} = ab^{-1}$ is in H .



Subgroup Tests

Two-Step Subgroup Test

Theorem: Let G be a group and H a nonempty subset of G . Then, H is a subgroup of G if ab is in H whenever a and b are in H (H is closed under multiplication), and a^{-1} is in H whenever a is in H (inverses).

Proof.

By the one-step subgroup test, it suffices to show that if a and b are in H , then ab^{-1} is in H . Since a^{-1} is in H whenever a is in H , we also know b^{-1} is in H . Thus, ab^{-1} is in H by closure under multiplication.



Subgroup Tests

Finite Subgroup Test

Theorem: Let H be a nonempty finite subset of a group G . Then, H is a subgroup of G if H is closed under the operation of G .

Proof.

By the two-step subgroup test, it suffices to show that if a is in H , then a^{-1} is in H . If $a = e$, then $a^{-1} = a$, and we are done.

If $a \neq e$, consider the sequence $a, a^2, \dots$. By closure, all of these elements belong to H . Since H is finite, not all of these elements will be distinct. Say $a^i = a^j$ and $i > j$. Then, $a^{i-j} = e$; and since $a \neq e$, this means that $i - j > 1$. Thus, $aa^{i-j-1} = a^{i-j} = e$ and therefore, $a^{i-j-1} = a^{-1}$. But, since $i - j - 1 \geq 1$, we know that a^{i-j-1} is in H , and we are done.



Examples of subgroups

A group G has always two subgroups at least:

- 1) The singleton $\{e\}$ where e is the neutral element of G .
- 2) The group G itself is a subgroup of G .

Inclusion of usual subgroups:

- 1) $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ by the addition as binary operation.
- 2) $\mathbb{Z}^* \leq \mathbb{Q}^* \leq \mathbb{R}^* \leq \mathbb{C}^*$ by the multiplication as binary operation,
- WARNING : Here , $\mathbb{Z}^* := \{-1, 1\}$

Permutation Groups

Definitions

Permutation of A : A *permutation* of a set A is a function from A to A that is both one-to-one and onto.

Permutation Group of a set A : A *permutation group* of a set A is a set of permutations of A that forms a group under function composition.

WORKED EXAMPLE 1:

For each of the following sets, determine whether the given operation is associative. Does the set contain an identity for the given operation? If it has an identity does every element have an inverse? Is the set a group under the given operation?

- (a) The even integers under addition.
- (b) The rationals under subtraction.
- (c) The rationals under multiplication.
- (d) The positive rationals under multiplication.
- (e) $\mathbb{Z}_6 \setminus \{[0]\}$ under multiplication.
- (f) $\mathbb{Z}_5$ under multiplication.
- (g) The cube roots of 1 under multiplication.

Solution W.E. 1

- (a) The even integers under addition.
- (b) The rationals under subtraction.
- (c) The rationals under multiplication.
- (d) The positive rationals under multiplication.
- (e) $\mathbb{Z}_6 \setminus \{[0]\}$ under multiplication.
- (f) $\mathbb{Z}_3$ under multiplication.
- (g) The cube roots of 1 under multiplication.

- (a) Associative; identity is 0; inverse of $2n$ is $-2n$; a group.
- (b) $(1-1)-1 \neq 1-(1-1)$ so not associative; no identity; not a group.
- (c) Associative; identity is 1; 0 has no inverse; not a group.
- (d) Associative; identity is 1; inverse of $\frac{r}{s}$ is $\frac{s}{r}$; a group.
- (e) Associative operation on $\mathbb{Z}_6$ but $\mathbb{Z}_6 \setminus \{[0]\}$ is not closed under the operation; for example, $[2][3] = [0]$.
- (f) Associative; identity is $[1]$; $[0]$ has no inverse; not a group.
- (g) Associative; identity is 1; inverse of $e^{2n\pi i/3}$ is $e^{-2n\pi i/3}$; a group.

WARNING: (d) : $\mathbb{Q}_+^*$

WORKED EXAMPLE 2 :

Which of the following sets are groups under the operation $*$ given?

- (a) Positive integers with $a * b = \max \{a, b\}$.
- (b) $\mathbb{Z}$ with $a * b = \min \{a, b\}$.
- (c) $\mathbb{R}$ with $a * b = a + b - ab$.
- (d) Positive integers with $a * b = \max \{a, b\} - \min \{a, b\}$.
- (e) $\mathbb{Z} \times \mathbb{Z}$ with $(a, b) * (c, d) = (a + c, b + d)$.
- (f) $\mathbb{R} \times \mathbb{R}$ with $(a, b) * (c, d) = (ac + bd, ad + bd)$.
- (g) $(\mathbb{R} \times \mathbb{R}) \setminus \{(0, 0)\}$ with $(a, b) * (c, d) = (ac - bd, ad + bc)$.

Solution W.E. 2

Question:

Which of the following sets are groups under the operation $*$ given?

- (a) Positive integers with $a * b = \max \{a, b\}$.
- (b) $\mathbb{Z}$ with $a * b = \min \{a, b\}$.
- (c) $\mathbb{R}$ with $a * b = a + b - ab$.
- (d) Positive integers with $a * b = \max \{a, b\} - \min \{a, b\}$.
- (e) $\mathbb{Z} \times \mathbb{Z}$ with $(a, b) * (c, d) = (a + c, b + d)$.
- (f) $\mathbb{R} \times \mathbb{R}$ with $(a, b) * (c, d) = (ac + bd, ad + bd)$.
- (g) $(\mathbb{R} \times \mathbb{R}) \setminus \{(0, 0)\}$ with $(a, b) * (c, d) = (ac - bd, ad + bc)$.

Answer:

- (a) No : 1 is the identity but $2 * a = 1$ implies $\max\{2, a\} = 1$ which is impossible.
- (b) No : there is no identity element (if there were such an element e then from $a * e = a$ we would have $\min\{a, e\} = a$ whence $a \leq e$ for all $a \in \mathbb{Z}$ and e would be the greatest element of $\mathbb{Z}$).
- (c) No : 0 is the identity but 1 has no inverse since $1 * b = 0$ implies $1 + b - b = 0$ which is nonsense.

- (d) No : the operation is not associative; for example, we have $2 * (3 * 4) = 2 * 1 = 1$ and $(2 * 3) * 4 = 1 * 4 = 3$.
- (e) Yes : $(0, 0)$ is the identity; $(-a, -b)$ is the inverse of (a, b) .
- (f) No: the operation is not associative; for example we have $((1, 2) * (2, 3)) * (2, 1) = (25, 17)$ and $(1, 2) * ((2, 3) * (2, 1)) = (17, 15)$.
- (g) Yes : $(1, 0)$ is the identity and the inverse of (a, b) is

$$\left(\frac{a}{a^2 + b^2}, -\frac{b}{a^2 + b^2} \right).$$

(Think about the non-zero complex numbers under multiplication!)

WORKED EXAMPLE 3

Question :

(a) Let G denote the group of non-zero rationals under multiplication.
Let

$$H = \{2^n \mid n \in \mathbb{Z}\}, \quad K = \left\{ \frac{1+2n}{1+2m} \mid n, m \in \mathbb{Z} \right\}.$$

Are H and K subgroups of G ?

(b) Show that $H = \{[0], [4], [8], [12]\}$ is a subgroup of $\mathbb{Z}_{16}$ under addition.

Answer :

(a) If $x, y \in H$ then $x = 2^n, y = 2^m$ for some $n, m \in \mathbb{Z}$ and so $xy^{-1} = 2^{n-m} \in H$. Hence H is a subgroup.

If $x, y \in K$ then for $n, m, r, s \in \mathbb{Z}$ we have

$$x = \frac{1+2n}{1+2m}, \quad y = \frac{1+2r}{1+2s}.$$

That K is a subgroup now follows from the fact that

$$xy^{-1} = \frac{1+2n}{1+2m} \cdot \frac{1+2s}{1+2r} = \frac{1+2(n+s+2ns)}{1+2(m+r+2mr)} \in K.$$

(b) Check that $[a], [b] \in H$ implies $[a] - [b] \in H$.

Exercise 1 :

(a) Let a and b be elements of a group G which commute. Prove that a and b^{-1} commute. Prove also that if x is any element of G then xax^{-1} and xbx^{-1} commute.

(b) Prove that a group G is abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.

(c) Prove that if G is a group such that $(ab)^2 = a^2b^2$ for all $a, b \in G$ then G is abelian.

(d) Let G be a group and suppose that $a^2 = 1$ for every $a \in G$. Prove that G is abelian.

(e) If G is a group and $x, y \in G$ prove by induction that, for every positive integer n , $(x^{-1}yx)^n = x^{-1}y^n x$.

(f) Let G be a group and suppose that $a, b \in G$ are such that $b^6 = 1$ and $ab = b^4a$. Prove that $b^3 = 1$ and that $ab = ba$.

Solution of the exercise 1

(a) Since $ab = ba$ we have $b^{-1}ab = b^{-1}ba = a$. Multiplying on the right

by b^{-1} we obtain $b^{-1}a = ab^{-1}$. Also, since a and b commute, we have

$$\begin{aligned}xax^{-1} \cdot xbx^{-1} &= xabx^{-1} \\ &= xba x^{-1} = xbx^{-1} \cdot xax^{-1}.\end{aligned}$$

(b) If G is abelian then $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$. Conversely, if this holds for all $a, b \in G$ then writing a^{-1} for a and b^{-1} for b we obtain $ba = ab$.

(c) If $(ab)^2 = a^2b^2$ then we have $abab = aabb$ and so, multiplying on the left by a^{-1} and on the right by b^{-1} we obtain $ba = ab$.

(d) If $a^2 = 1$ for all $a \in G$ then given any $x, y \in G$ we have $(xy)^2 = 1$, $x^2 = 1$ and $y^2 = 1$. Consequently $(xy)^2 = x^2y^2$ and the result follows by (c).

(e) The result is trivial for $n = 1$. For the inductive step, suppose that $(x^{-1}yx)^n = x^{-1}y^n x$; then

$$(x^{-1}yx)^{n+1} = (x^{-1}yx)^n x^{-1}yx = x^{-1}y^n x x^{-1}yx = x^{-1}y^{n+1}x.$$

(f) We are given that $b^6 = 1$ and $ab = b^4a$. From the second of these and (e) we obtain $b = a^{-1}b^4a = (a^{-1}ba)^4$. Thus

$$b^3 = (a^{-1}ba)^{12} = a^{-1}b^{12}a = a^{-1}1a = 1,$$

and $ab = b^4a = b^3ba = ba$.

Question: Which of the following groups are abelian?

*

- 1) The non-zero elements of $\mathbb{Z}_{11}$ under multiplication.
- 2) The group of bijections of $E = \{1, 2, 3\}$ to itself under composition of mappings.
- 3) The group of bijections of $E = \{1, 2\}$ to itself under composition of mappings.
- 4) The even integers under addition.
- 5) $\mathbb{R}$ under addition.

Answer : You have to prove it ...

- 1) Abelian
- 2) Not abelian
- 3) Abelian
- 4) Abelian
- 5) Abelian

Exercise 2 and its solution

Problem :

Define a binary operation $*$ on $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ by

$$a * b = \begin{cases} ab & \text{if } a > 0; \\ \frac{a}{b} & \text{if } a < 0. \end{cases}$$

Is $\mathbb{R}^*$ a group under $*$?

Solution :

The operation $*$ is associative. To see this, we must check several cases. For example, suppose that $a < 0$ and $b < 0$; then we have

$$(a * b) * c = \frac{a}{b} * c = \frac{ac}{b} \quad \text{and} \quad a * (b * c) = a * \frac{b}{c} = \frac{ac}{b}.$$

Similar arguments in the cases $a < 0, b > 0$ and $a > 0, b < 0$ and $a > 0, b > 0$ complete the proof of associativity.

1 is the identity element. In fact $1 * a = 1a = a$; and for $a > 0$ we have $a * 1 = a1 = a$, while for $a < 0$ we have

$$a * 1 = \frac{a}{1} = a.$$

If $a > 0$ then $\frac{1}{a}$ is its inverse; for we have

$$a * \frac{1}{a} = a \frac{1}{a} = 1, \quad \frac{1}{a} * a = \frac{1}{a} a = 1.$$

If $a < 0$ then a is its own inverse; for we have

$$a * a = \frac{a}{a} = 1.$$

Hence the operation $*$ makes $\mathbb{R} \setminus \{0\}$ into a group.

Exercise 3

Exercise 3. 1. Show that the following sets G endowed with the given laws $\star$ form groups. Exhibit the neutral element, and the inverse of $x \in G$.

- (a) $G = \mathbb{Z}$, $\star$ = the addition of numbers;
- (b) $G = \mathbb{Q}^*$ (the set of non-zero rationals), $\star$ = the multiplication of numbers;
- (c) $G = \mathbb{Q}^{+*}$ (the set of positive rationals), $\star$ = the multiplication of numbers;
- (d) $G = \mathbb{R}$, $\star$ = the addition of numbers;
- (e) $G = \mathbb{R}^*$, $\star$ = the multiplication of numbers;
- (f) $G = \mathbb{R}^{+*}$, $\star$ = the multiplication of numbers;
- (g) $G = \mathbb{C}$, $\star$ = the addition of numbers;
- (h) $G = \mathbb{C}^*$, $\star$ = the multiplication of numbers;
- (i) $G = \{z \in \mathbb{C}, |z| = 1\}$, $\star$ = the multiplication of numbers;
- (j) $G = \{e^{i\frac{2\pi k}{n}}, k = 0, 1, \dots, n-1\}$, $\star$ = the multiplication of numbers (n is a fixed integer);
- (k) G = the set of bijections of a non-empty set E , $\star = \circ$ (the composition of functions);

Solution of exercise 3 (a,b, c, d,e,f, g, h)

Correction 3. 1. (a) For $G = \mathbb{Z}$ with $\star = +$ (the addition of numbers), the neutral element of the group law is $e = 0$, and the inverse of $x \in \mathbb{Z}$ is $-x \in \mathbb{Z}$.

(b) For $G = \mathbb{Q}^*$ (the set of non-zero rationals) with $\star = \cdot$ (the multiplication of numbers), the neutral element of the group law is $e = 1$. The inverse of $\frac{p}{q} \in \mathbb{Q}^*$, is $\frac{q}{p} \in \mathbb{Q}^*$.

(c) For $G = \mathbb{Q}^{+*}$ (the set of non-negative rationals) with $\star = \cdot$, one uses that the product of two positive numbers is positive, and that $\frac{q}{p} > 0$ whenever $\frac{p}{q} > 0$.

(d) For $G = \mathbb{R}$ with $\star = +$, the neutral element of the group law is $e = 0$, and the inverse of $x \in \mathbb{R}$ is $-x \in \mathbb{R}$.

(e) For $G = \mathbb{R}^*$ with $\star = \cdot$, the neutral element of the group law is $e = 1$, and the inverse of $x \in \mathbb{R}^*$ is $\frac{1}{x} \in \mathbb{R}$.

(f) For $G = \mathbb{R}^{+*}$ with $\star = \cdot$, one uses in addition to the previous item that the set of positive numbers is stable by product and inverse.

(g) For $G = \mathbb{C}$ with $\star = +$, the neutral element is 0, and the inverse of $x = a + ib \in \mathbb{C}$ is $-x = -a - ib \in \mathbb{C}$.

(h) For $G = \mathbb{C}^*$ with $\star = \cdot$, the neutral element is $e = 1 \in \mathbb{C}$, and the inverse of $x = a + ib \in \mathbb{C}$ is

$$\frac{1}{x} = \frac{1}{a + ib} = \frac{a}{a^2 + b^2} - i \frac{b}{a^2 + b^2}.$$

Solution of exercise 3 (i, j, k)

- (i) For $G = \{z \in \mathbb{C}, |z| = 1\}$ with $\star = \cdot$, one uses that $|z_1 z_2| = |z_1| \cdot |z_2|$, hence the product of two complex numbers of module 1 is a complex number of module 1. The neutral element is $e = 1$, and the inverse of $x = e^{i\theta} \in G$ is $e^{-i\theta} \in G$.
- (j) For $G = \{e^{i\frac{2\pi k}{n}}, k = 0, 1, \dots, n-1\}$ with $\star = \cdot$, where n is a fixed integer, one uses that $e^{i\frac{2\pi k_1}{n}} \cdot e^{i\frac{2\pi k_2}{n}} = e^{i\frac{2\pi(k_1+k_2)}{n}}$, and

$$\frac{1}{e^{i\frac{2\pi k}{n}}} = e^{-i\frac{2\pi k}{n}}.$$

- (k) For $G =$ the set of bijections of a non-empty set E , with $\star = \circ$ (the composition of functions), one uses that the composition of two bijections is a bijection, and that a map which is injective and surjective admits an inverse map. The neutral element is the identity map. The inverse of a bijection f is commonly denoted by f^{-1} but has usually nothing to do with the map $\frac{1}{f}$ (if this ever makes sense);

Exercise 4

Exercise 4. Say for which reason(s) the following operations $\star$ do not endow the given sets G with a group structure.

- (a) $G = \mathbb{N}$, $\star =$ the addition of numbers;
- (b) $G = \mathbb{N}^*$, $\star =$ the multiplication of numbers;
- (c) $G = \mathbb{R}$, $\star =$ the multiplication of numbers.

GROUP MORPHISM

Definition :

Let G and H be groups, and $f:G \rightarrow H$.

Then f is a morphism if:

For all $x, y \in G$, $f(xy) = f(x) f(y)$.

If a morphism is also a bijection, then it is called an *isomorphism*.

If $G=H$, it is called endomorphism.

A bijective endomorphism is called automorphism.

Question:

Show that :

a) $f(e) = e$, where e is the neutral element of G .

b) $f(x^{-1}) = (f(x))^{-1}$, where $x \in G$.

EXAMPLES of MORPHISMS

- Examples :

1) Id_G is an automorphism of the group G .

2) For a group G and for any $a \in G$,

$$f: G \rightarrow G$$

defined by

$$f(x) = axa^{-1},$$

is an automorphism.

Kernel and image of a group morphism

- There are two important subsets associated with every group morphism:

$$f: G \rightarrow H$$

namely the kernel and image of f , defined respectively by

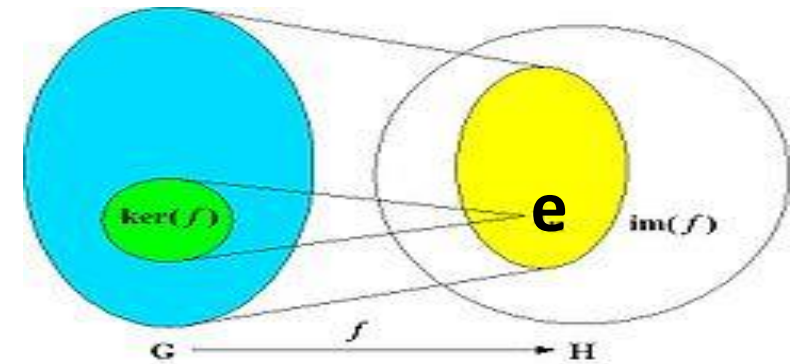
$\text{Ker } f := \{x \in G, f(x) = e\} = f^{-1}(\{e\})$, where e is the neutral element of H .

$\text{Im } f := \{f(x) \in H, x \in G\} = f(G)$.

Theorem:

1) $\text{Ker } f$ is a subgroup of G .

2) $\text{Im } f$ is a subgroup of H .



Proof : In fact : a) inverse image of a subgroup of H is a subgroup of G .

b) direct image of a subgroup of G is a subgroup of H .

preimage (or inverse image)of a group morphism:

- One proof...

Prove that if $f : G \rightarrow H$ is a group homomorphism and $K \leq H$ then the preimage of K , defined as $f^{-1}(K) = \{g \in G | f(g) \in K\}$, is a subgroup of G .

PROOF:

1) Note that $f^{-1}(K)$ is nonempty, since $f(e_G) = e_H$ and $e_H \in K$. So $e_G \in f^{-1}(K)$.

2) Let $x, y \in f^{-1}(K)$. So $f(x), f(y) \in K$ by the definition of the preimage of K . Then $f(xy^{-1}) = f(x)f(y^{-1}) = f(x)f(y)^{-1} \in K$ because $f(x), f(y) \in K$ and K is closed under multiplication and inverses. Hence $xy^{-1} \in f^{-1}(K)$.

We conclude that $f^{-1}(K) \leq G$.

$$\begin{aligned} \text{id} : G &\rightarrow G & \text{id}(x) &= x \\ i : H &\rightarrow G & i(x) &= x \end{aligned}$$

Examples of group morphisms:

Example 1

$f : \mathbb{Z} \rightarrow \mathbb{Z}$ with $a \mapsto 3a$ is a **group homomorphism** between the **group** $(\mathbb{Z}, +)$ and its **subgroup** $(\{3n \mid n \in \mathbb{Z}\}, +)$, since

$$f(a + b) = 3(a + b) = 3a + 3b = f(a) + f(b)$$

for all integers $a, b \in \mathbb{Z}$.

Example 2

For **real numbers** $x \in \mathbb{R}$, the **exponential function** is a **group homomorphism** between the groups $(\mathbb{R}, +)$ and $(\mathbb{R}_+^*, \cdot)$, since

$$\exp(x + y) = \exp(x) \cdot \exp(y)$$

for all $x, y \in \mathbb{R}$. This is also known as the **functional equation of the exponential function**.

Problem (with solution)

1)

Let $G = \{x \in \mathbb{R} \mid x^2 < 1\}$. Show that if $x, y \in G$ then

$$\frac{x+y}{1+xy} \in G.$$

Prove that G is a group under the law of composition

$$x * y = \frac{x+y}{1+xy}.$$

1)

$$\left(\frac{x+y}{1+xy}\right)^2 < 1 \Leftrightarrow x^2 + 2xy + y^2 < 1 + 2xy + x^2y^2$$

$$\Leftrightarrow 1 + x^2y^2 - x^2 - y^2 > 0$$

$$\Leftrightarrow (1-x^2)(1-y^2) > 0.$$

Consequently, if $x, y \in G$ then

$$\frac{x+y}{1+xy} \in G.$$

For $x * y = \frac{x+y}{1+xy}$ it is readily seen that

$$x * (y * z) = \frac{x + y + z + xyz}{1 + xy + xz + yz} = (x * y) * z.$$

Thus $*$ is associative. Clearly, $*$ is commutative; and since $x * 0 = x$ it follows that 0 is the identity element. Finally, $x * (-x) = 0$ shows that $-x$ is the inverse of x under $*$. Hence G is a group.

2)

By considering the mapping $f : G \rightarrow \mathbb{R}$ described by

$$f(x) = \log \frac{1+x}{1-x},$$

prove that the group G and the additive group $\mathbb{R}$ are isomorphic.

• 2)

With $f(x) = \log \frac{1+x}{1-x}$ we have

$$f(x * y) = \log \frac{1 + \frac{x+y}{1+xy}}{\frac{x+y}{1+xy}} = \log \frac{1+xy+x+y}{1+xy-x-y}$$

$$f(x) + f(y) = \log \left(\frac{1+x}{1-x} \cdot \frac{1+y}{1-y} \right) = \log \frac{1+xy+x+y}{1+xy-x-y}$$

and so f is a morphism.

That f is injective follows from the fact that

$$f(x) = 0 \Rightarrow \log \frac{1+x}{1-x} = 0 \Rightarrow \frac{1+x}{1-x} = 1 \Rightarrow x = 0.$$

To see that f is also surjective, we observe that $x \rightarrow \frac{1+x}{1-x}$ maps $] -1, 1[$ onto

$\mathbb{R}_+$, and $x \rightarrow \log x$ maps $\mathbb{R}_+$ onto $\mathbb{R}$. Hence f is an isomorphism.

RINGS : definition

- A ring R is a non-empty set with two laws of composition, written as addition and multiplication, such that
- (1) R is an abelian group under addition;
- (2) the multiplication is associative;
- (3) these laws are linked by the **distributive** laws:
- $(\forall x, y, z \in R) \quad (x + y)z = xz + yz \text{ and } z(x + y) = zx + zy$
- A ring $[R; +, \cdot]$ that has a multiplicative identity is called a **ring with unity**.
- The multiplicative identity itself is called the unity of the ring.
- More formally, if there exists an element $1 \in R$, such that for all $x \in R$, $x \cdot 1 = 1 \cdot x = x$, then R is called a ***ring with unity***.

EXAMPLE : $\mathbb{Z}$: « *The Lord of the Rings* »



$[\mathbb{Z}, +, \cdot]$ is a ring, where $+$ and $\cdot$ stand for regular addition and multiplication on $\mathbb{Z}$. We already know that $[\mathbb{Z}, +, \cdot]$ is an abelian group, so we need only check parts 2 and 3 of the definition of a ring. From elementary algebra, we know that :
the associative law under multiplication
and the distributive laws are true for $[\mathbb{Z}, +, \cdot]$.
This is our main example of an infinite ring,

EXAMPLE 2 : $\mathbb{Z}/n\mathbb{Z}$: The Ring of integers mod n

$$\mathbb{Z}/n\mathbb{Z} := \{ \overline{0}, \overline{1}, \overline{2}, \dots, \overline{n-1} \}$$

Prove it...

This example is our main example of finite rings of different orders.

Commutative Ring

- When the multiplication is commutative we say that R is a commutative ring.

Zero divisors and Integral domain

- The additive identity element 0 is called the **zero of the ring**.
- In a ring it is possible to have $xy = 0$ with $x \neq 0$ and $y \neq 0$.

Such elements are called **zero divisors**. A commutative ring with a 1 which has no zero divisors is called an **integral domain**.

- **Example:**

- In $\mathbb{Z}/12\mathbb{Z}$, we have : $\bar{2} \cdot \bar{6} = \bar{0}$, with $\bar{2} \neq 0$ and $\bar{6} \neq 0$.
 $\bar{2}$ and $\bar{6}$ are zero divisors, and $\mathbb{Z}/12\mathbb{Z}$ is **not** an integral domain.